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Multistage Cemented Sleeves in Vertical Wells - Lesson Learned

Khamis Salim Al Hosni, Petroleum Development Oman, Muscat, Oman; Stevanus Dwi kurniadi, Schlumberger, Muscat, Oman; Benjamin Mowad, Petroleum Development Oman, Muscat, Oman; Tausif Khizar Ahmed, Schlumberger, Muscat, Oman; Ali Al Ghaiti, Ghusun Al Aamri, and Fathy Sulaiman, Petroleum Development Oman, Muscat, Oman

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Abstract

Stacked, tight gas reservoirs in central Oman require production enhancement from hydraulic fracturing to realize economic gas rates. Typically, plug and perf techniques are utilized for this purpose. However, completion alternative technology: Multistage frac sleeves (MSF) with closeable option were considered in a three well trial to gauge effectiveness in shutting off watered out zones. Secondary advantage of MSF would be operational efficiency gains by removal of wireline interventions entirely.

Cemented sleeve completions in vertical wells are not common in today's oilfields globally. To remove as much uncertainty as practical, well candidates were chosen in fields/areas with historically high fracturing success rates (proppant placement as designed). The drilling requirements and operation were generally similar to conventional completions. The fracture design was adjusted to be more conservative than standard until frac success was established. Upon completion of the frac operation, ball seats were milled to allow full-bore access in subsequent interventions and ability to reclose any sleeve

The result of the MSF trials was encouraging in this first deployment. The drilling and completion processes were smooth and without issues. Cased hole operations saw the successful opening of the toe initiation valve in all cases. The proppant placement ratio was even higher as compared to offset wells. Post frac production assessment shows comparable initial results, although it still require long-term testing for full validation of the MSF as a differentiator. Operational efficiencies were not able to achieve a significantly lower stage cycle as expected despite having a similar cycle per stage as plug and perf. This was due to several challenges during execution, such as possible debris or proppant between the ball and seat which meant a leak path to below zones during the frac. A screen out was experienced, high near-wellbore friction was common and so on. These unexpected hurdles required additional interventions including coiled tubing cleanouts, added perforations, and well flowbacks.in future deployments, MSFs will be required in multiple sand bodies to ensure the better zonal coverage, this was not achieved per the radioactive tracer results. Mastering the operation and adjusting the requirement during the ball drop operation will allow reduction of operation time as well as avoiding over flush.

In this paper we share the journey and lessons learned of an uncommon application of a common technology (MSF). Unlike application in horizontal wells, a vertical well with MSF provides a different point of view on the technology capability. Challenges and solutions will be discussed to enrich Middle East fracturing experience.

Introduction

Oman has established itself as a regional leader in the development of tight gas resources, driven by the strategic application of advanced drilling and stimulation technologies. The country has successfully leveraged hydraulic fracturing, multi-stage stimulation, and horizontal drilling to unlock production from low-permeability reservoirs that were previously considered uneconomical.

The journey began in the late 1990s, when Petroleum Development Oman (PDO) pioneered the use of hydraulic fracturing in the country. These early efforts laid the foundation for what would become a sustained and evolving approach to unconventional gas development. Since then, the practice has been adopted and further advanced by several other operators.

Across Oman, and particularly within the Ghaba Salt Basin, tight gas development has primarily targeted two key clastic reservoirs: the Barik (B) Formation and the Miqrat (M) Formation. Both formations are part of the Haima Supergroup, which comprises a thick sequence of siliciclastic sediments deposited during the Cambrian to Early Silurian periods.

The B Formation is extensively distributed across the central and northern regions of the Sultanate. It is predominantly interpreted as a braided delta complex, transitioning into a shallow marine depositional system toward the north. This formation is a significant tight gas reservoir, encountered at depths ranging from 4,300 to 4,600 meters. It is subdivided into nine distinct reservoir units, each of which typically requires hydraulic stimulation to achieve commercial gas production.

The M Formation represents a complex and heterogeneous clastic sequence deposited during the Cambrian period. It is interpreted as having formed in three primary end-member depositional environments: sabkha, alluvial plain, and playa lake. However, the mixed sedimentological character of many beds suggests that the actual depositional setting at any given time was transitional, reflecting dynamic shifts between these environments. Stratigraphically, the MFormation is subdivided into three major units: Lower, Middle, and Upper M. Of these, the Middle M is the principal target for hydraulic fracturing due to its favorable reservoir properties. This unit is further divided into six sub-units, each exhibiting varying degrees of reservoir quality and stress conditions (Ibrahim, 2018).

Hydraulic Fracturing Overview

The B and M reservoirs are produced commingled in the field; a strategy adopted to optimize recovery from both formations with differing reservoir characteristics. Since the onset of production in 1999 from the Barik reservoir—and the subsequent initiation of commingled production with the Miqrat reservoir in 2001—hydraulic fracturing has remained a core component of the field's development strategy.

Hydraulic fracturing operations in the B Formation are tailored to the geological variability across the field. Depending on the well location, reservoir quality, and stress regime, between five and nine fracturing stages are commonly executed within this interval. These multi-stage treatments are designed to maximize reservoir contact, improve fracture conductivity, and enhance gas recovery from the low-permeability sandstone units. In the Middle M formation typically two to four stages frac planned, depending on the well location, formation thickness, and geomechanical properties. These treatments are designed to enhance gas recovery from the tight, low-permeability sandstones that dominate the unit (Ibrahim, 2018).

Over the years, significant heterogeneity in reservoir permeability and differential depletion between the two formations have posed increasing challenges to production performance (Ibrahim, 2018). These complexities have necessitated a continuous evolution in the hydraulic fracturing approach, including:

- Adjustments in the number of fracturing stages per well, tailored to reservoir layer and stress contrasts.
- Optimization of proppant size and type, to improve fracture conductivity.
- Refinement of pad volumes and fluid systems, to enhance fracture initiation and propagation in low-permeability zones.
- Adaptation of operational methodologies, including real-time monitoring and post-frac diagnostics, to improve treatment effectiveness and reduce operational risks.
- Completion type includes vertical and horizontal well.

This adaptive fracturing strategy has been critical in maintaining production levels and extending the economic life of the field, despite the natural decline observed in the Barik reservoir and the variable contribution from the Miqrat formation.

Hydraulic Fracturing Challenges

With the current standard approach of hydraulic fracturing operation using plug and per methodology, there are several challenges that are observed across the field.

- Differential depletion across the sand layer.

  After 30+ year of production and different permeability across the sand layer, the current state of reservoir pressure is quite varying. The reservoir pressure has been reduced from 580 bars to around 100 bars in some zones. The difference depletion between each sand layer can be seen in Figure 1 below.

  In fields experiencing uneven or differential depletion, several operational and mechanical issues have begun to emerge, particularly during and after hydraulic fracturing. These challenges can significantly impact performance, intervention efficiency, and overall recovery. Key issues include:

  - Difficulty in Plug Milling and Well Cleanout Post-Fracturing

    Excessive fluid leak-off into depleted zones results in little to no fluid returns during plug milling operations. Without sufficient fluid circulation, cuttings generated during milling cannot be effectively transported to the surface, leading to accumulation in the wellbore and potential equipment damage or stuck pipe incidents.
  - Increased Stress Contrast Between Reservoir Layers

    Differential depletion creates a higher stress contrast between sand-rich reservoir zones and overlying or underlying shale barriers. This increased contrast can lead to stronger fracture containment within the sand body, limiting vertical fracture growth. While this may help avoid unwanted fracture propagation, it can also reduce the zone coverage if not properly accounted for in the design.
  - Casing Deformation Despite Vertical Well Geometry

    Some wells have experienced casing deformation even though they are drilled vertically. In severe cases, this deformation restricts access to certain zones, preventing plug milling and post-frac cleanout. With current plug-and-perf systems, there is a risk that some intervals may remain untreated or unproductive due to inaccessibility.
  - Liquid Loading During Production and Maintenance

    Depleted zones often suffer from poor reservoir pressure support, leading to liquid loading in the wellbore. This can hinder hydrocarbon flow, reduce production rates, and increase the need for artificial lift or frequent well interventions to remove accumulated fluids.
- Less than optimum operational efficiency during frac operation

The plug-and-perf (PnP) method, while widely used in hydraulic fracturing, introduces several operational inefficiencies—especially in wells affected by casing deformation or access restrictions. One of the key challenges is the additional time required for setting plugs and performing perforations, which often exceeds the actual fracturing time per stage. These inefficiencies highlight the potential benefits of alternative completion strategies, such as sliding sleeve systems, coiled tubing-conveyed fracturing, or simul-frac operations, which can reduce NPT and improve operational continuity.

- Water conformance

  During the life of the well, several sand bodies—particularly near the crest—have shown high water production. This is a common issue in mature or structurally complex reservoirs, where water conning or breakthrough can occur due to pressure imbalance or fracture connectivity. If the water-producing zone is located within the B formation, isolating it with a permanent plug could unintentionally shut off the entire M formation, which may still be contributing valuable hydrocarbon production. isolating individual fractured zones that produce water is technically challenging. The available options are Casing patches: These can be used to cover the perforated interval, but they are often temporary and may not provide a complete seal.

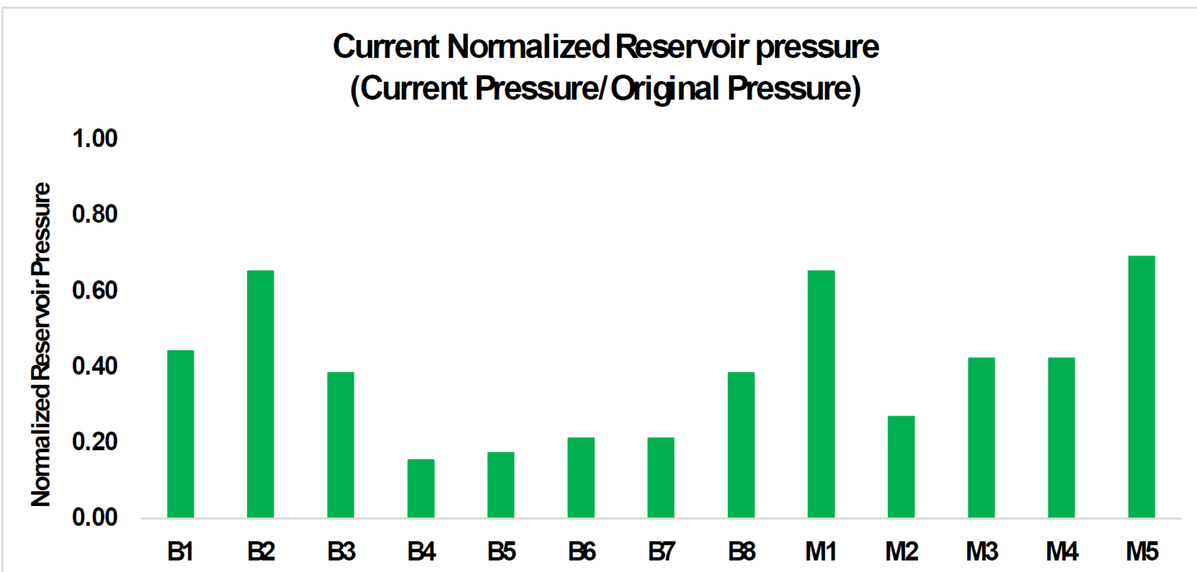

Figure 1—Normalized current reservoir pressure across the B and M formations (Modified After Al Bahri, 2022)

These inefficiencies and challenges highlight the potential benefits of alternative completion strategies, such as multistage stimulation with open holes or cemented sleeve, coiled tubing-conveyed fracturing, which can reduce NPT and improve operational continuity.

Overview of MSF in the Industry and Oman

After a thorough evaluation of alternative completion systems, the multistage cemented sleeve has been selected for trial implementation in Oman. This decision was made after comparing various multistage fracturing (MSF) systems, including the open hole MSF approach.

The cemented sleeve MSF system was chosen due to several key advantages over the open hole (uncemented) MSF system:

- Elimination of Open Hole Packers: Unlike the open hole MSF system, the cemented MSF does not rely on open hole packers, which are often a critical factor in the go/no-go decision for deployment.

Open hole packers can pose a higher risk of getting stuck or experiencing high drag during the running-in-hole operation.

- Improved Zonal Isolation: In the cemented MSF system, zonal isolation is achieved through the cement behind the casing, rather than relying on mechanical packers. This provides a more robust and reliable isolation method, especially in challenging wellbore conditions.
- Reduced Dependence on Wellbore Quality: The cemented MSF system is less affected by washouts or irregularities in the wellbore. In contrast, the open hole MSF system requires a well-gauged and uniform borehole to ensure effective operation, which can be difficult to achieve in certain formations.

On the other hand, the cemented MSF system does come with its own limitations, which have been carefully considered prior to its selection (Rudoplu, 2024). These include:

- Limited Flexibility in Fracture Placement: Once the cemented liner is in place, there is minimal flexibility to adjust the location of fracture stages. This can be a constraint in formations where precise targeting of fracture zones is critical.
- Risk of Cement Bond Failure: The effectiveness of zonal isolation in cemented MSF systems depends heavily on the quality of the cement bond. Any failure in the cement bond can compromise isolation between zones, potentially affecting stimulation efficiency and well integrity.
- Stage Limitations Under High Differential Pressures: Cemented MSF systems may be limited in the number of stages that can be effectively deployed, especially in wells requiring high differential pressure. This could restrict their application in more complex or high-pressure environments.

Toe initiation valve

The Toe Initiator Valve (TIV) is a pressure-activated port designed to establish initial communication between the wellbore and the reservoir. It serves as the first-stage initiator in a multi-stage hydraulic fracturing operation, particularly in horizontal wells. Several types of TIV mechanisms are available in the market, each with distinct operational principles:

- **Wet Shoe System**. This method utilizes a rupture disc integrated into the cementing plug system. After the cementing process is completed, a second dart is pumped downhole to apply pressure on the rupture disc. Once the disc bursts, it opens a flow path through the shoe, allowing fluid to enter the formation and initiate the first fracture.
- **Toe Sub**. A toe sub is a specialized tool installed at the toe of the well, equipped with one or more rupture discs. These discs are designed to burst at predetermined pressures, creating a flow path to the reservoir.
- **Toe Sleeve**. The toe sleeve consists of a sleeve and piston assembly. When the internal pressure reaches the designed threshold, a rupture disc bursts, triggering the piston to shift and open the sleeve. This action exposes ports that allow fluid to flow into the formation.

Cemented frac sleeve

The cemented frac sleeve is a completion mechanism designed to replace traditional perforation methods by providing port to establish communication between the wellbore and the reservoir. These sleeves are deployed in a closed position, allowing for seamless passage of cement and wiper darts without the risk of premature activation (Rodoplu, 2024).

To ensure minimal pressure drop across the ports during stimulation, the flow area of the frac sleeve is engineered to be equal to or greater than the internal flow area of the wellbore. This design consideration is critical for maintaining low frictional losses during fracturing operations.

Frac sleeves can be mechanically activated using coiled tubing or hydraulically activated via a ball-drop system. Their slim outer diameter will reduce the stuck event in challenging wellbore conditions. In both cemented and openhole multistage fracturing (MSF) systems, frac sleeves can be configured in two primary designs: single entry and multi entry port.

In Single-Entry Port Design, the configuration features a single, fully open port that provides unrestricted flow to the formation. In limited-Entry Multi-Port Design, each port is intentionally designed with a restricted flow area to create a pressure drop across multiple entry points. This encourages more uniform fluid distribution across the stage. Over time, erosion occurs at the port openings due to high-velocity fluid flow, gradually enlarging the ports and enhancing flow capacity.

MSF Design and Plan

The deployment of Multi-Stage Fracturing (MSF) technology in Oman is the result of a comprehensive and systematic review process, which carefully evaluated the application scope, system requirements, and engineering design. This initiative represents a tailored approach to stimulation, specifically adapted to the geological and operational context of the region.

In contrast to conventional MSF applications commonly used in horizontal wells across other markets, the Omani implementation is uniquely designed for vertical wells, with each well accommodating up to eight fracturing stages. This configuration is strategically developed to target the B and M formations, which are characterized by their layered lithology and variable reservoir properties.

MSF Objective

The primary objectives behind the deployment of Multi-Stage Fracturing (MSF) technology in Oman are centered around enhancing well performance, improving operational efficiency, and increasing system flexibility. These objectives include:

- Achieving Comparable Productivity to Plug-and-Perf Completions. The MSF system is designed to deliver well productivity and reservoir contact comparable to that of the conventional plug-and-perf methodology, ensuring effective stimulation across all targeted zones.
- Optimizing Fracturing Unit Efficiency. By streamlining the stimulation process and reducing the need for repeated interventions, the MSF system aims to maximize the operational efficiency of the fracturing spread, thereby reducing overall treatment time and cost.
- Minimizing Milling Operations. The system is engineered to reduce or eliminate the need for post-fracturing milling operations, which are typically required in plug-and-perf completions. This contributes to shorter well completion timelines and lower mechanical risk.
- Enabling Independent Stage Control. One of the key advantages of the MSF system is the ability to selectively open and close individual stages. This functionality allows for greater control over stimulation placement, re-stimulation opportunities, and production management.
- Mitigating the Impact of Casing Deformation. In the event of casing deformation, the MSF system is designed to maintain functionality and minimize operational disruption. This resilience enhances the reliability of the completion system in challenging downhole environments.
- Cost efficiency. With the operation efficiency and less risk on the intervention, the cost efficiency will be the advantages of the MSF completion system.

MSS Requirements

To align with the operational objectives and ensure reliable performance in the challenging downhole environments of Oman, the Multi-Stage Fracturing (MSF) system selected for Petroleum Development

Oman (PDO) must meet a set of stringent technical requirements. These specifications are designed to ensure durability, flexibility, and efficiency throughout the stimulation and production lifecycle.

The key system requirements are as follows:

- High-Pressure Rating and Corrosion Resistance. The system must be rated for a minimum differential pressure of 15,000 psi, constructed using 13CrS (13% chromium stainless steel) material to ensure corrosion resistance in high-pressure, high-temperature (HPHT) and sour service conditions.
- Reliable First-Stage Initiation Without Perforation. The first stage must be initiated without the need for perforation, requiring a high-reliability initiation valve capable of consistent activation under downhole conditions.
- Dual-Activation Sleeve Design. The system must support interchangeable sleeves that can be activated either by ball drop or coiled tubing, providing operational flexibility depending on well conditions and intervention strategy.
- Reclosable Sleeves. Each stage sleeve must be reclosable, allowing for selective shut-off or re-stimulation of individual zones during the life of the well.
- Full-Bore Flow Area. The internal flow area of the sleeves must be equivalent to the nominal casing ID, ensuring minimal flow restriction and facilitating post-frac production and intervention operations.
- Limited Entry Multi-Sleeve Capability. The system must support limited entry fracturing through multiple sleeves per stage, enabling even distribution of treatment fluids across perforated clusters and improving stimulation efficiency.
- Dissolvable Ball Technology. The use of dissolvable balls is required to eliminate the need for post-frac retrieval or milling, reducing operational complexity and risk.

MSF Design

The deployment of the Multi-Stage Fracturing (MSF) system in Petroleum Development Oman (PDO) is structured to follow a phased approach, allowing for a learning curve in its application. This deployment is divided into several trial phases, each with specific objectives, as illustrated in Figure 2 below.

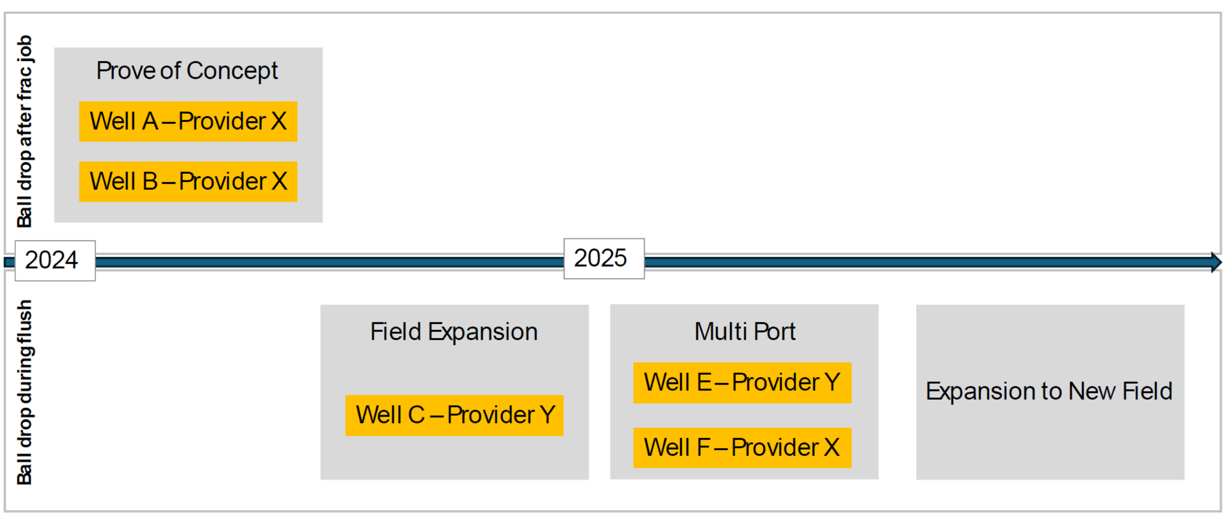

Figure 2—Trial plan of the MSF in Petroleum Development Oman

Since the application of MSF in vertical wells is relatively rare, the trials are designed to achieve several key objectives:

- Proof of Concept
  The primary goal of this phase is to validate the feasibility of the system. This includes verifying the successful installation of the MSF system, testing the opening of the Tubing Isolation Valve (TIV) and assessing the impact of cemented MSF sleeves on the fracturing operation.
- Evaluation of Ball Drop Methods and Overflush Risk
  There are multiple methodologies for deploying the ball used to open the sleeves. The two most common are on-the-fly ball drop during the flush stage and post-fracturing ball drop after the stage is completed. Each method has its own advantages and drawbacks, particularly concerning the risk of overflushing the previous stage. This trial phase will evaluate both methods to determine their impact on operational efficiency and production outcomes.
- Use of Multi-Stage Ports to Enhance Zone Coverage
  Given the laminated nature of the reservoir and its level of depletion, a single-port system may not provide adequate coverage across the entire reservoir section. This phase will trial multiple ports to assess their effectiveness in improving zone coverage and maximizing reservoir contact.
- Impact of Cemented MSF on Breakdown Across Different Fields
  Theill be conducted in a field and formation that is well understood and commonly stimulated. This will help establish a baseline for system performance, including breakdown pressure and job placement. Subsequent trials in different fields will evaluate how the cemented MSF system performs under varying geological conditions.

To add typical design, picture of the completion, etc

MSS surveillance plan

A radioactive (RA) tracer is planned to be used as part of the evaluation process for the cemented Multi-Stage Fracturing (MSF) system. The tracer will help assess the zone coverage of each stage, particularly since the treatment is designed to stimulate multiple sand layers. Additionally, the RA tracer will be used to verify whether the dissolvable ball effectively isolates each zone, ensuring proper stage separation and treatment placement.

Technology Deployment and Result

Well A

Well A was planned to be completed with 8 stages with single port sleeves - 3 sleeves in the M Reservoir and 5 in the B reservoir. The integrity test and opening of the TIV were performed 81 days prior to the start of the frac phase, and the frac tree was installed shortly after. A modular tank was also installed at the wellsite and filled prior to the start of the frac phase to improve efficiency between stages. Table 1 shows the stimulation summary of the stages. Overall, breakdown was achieved in all stages except stage 6 (B-3). After 6 injection attempts, contingency perforations were performed on this stage. The channel fracturing technique was used at the beginning of every stage as means to manage the net pressure help with proppant placement. Once pressure trend stabilized, the switch to conventional proppant stages was made in order to meet the target tonnage. The Step-Down Tests (SDT) conducted prior to the main fracturing treatments confirmed the high near wellbore friction (~1500 – 3000 psi), attributed to increased tortuosity. This also placed a proppant acceptance limit, causing near wellbore screenouts in stage 3&4 (Figure 3). Initially, more 30/50 ISP was planned to be pumped during the earlier proppant stages to mitigate the risk of early bridging. However, as the pressures stabilized earlier than expected on the stages, the additional 30/50 was not pumped.

Table 1—Well A Stimulation Summary

Stage	Zone	Distance	Contingency Perf?	Frac Plan (MT)	Max PPA plan	Actual placed (MT)	Max PPA actual	Fluids	Actual PAD%	Avg Rate (m3/min)	Avg Frac Press (kPa)
1	M-5 (TIV)		N	60	6	63	5	40# & 35# XL gel	24.7%	4.2	58,281
2	M-3&4	17.2	N	116	7	127.5	7	40# & 35# XL gel	31.8%	4.8	54,112
3	M-1&2	27.7	N	58.3	6	41.7 (SO)	4	40# & 35# XL gel	30.5%	4.3	56,166
4	B-5	366.3	N	43.3	6	43.3 (SO)	4	35# XL gel	32%	4.2	52,330
5	B-4	11.2	N	30	5	34.5	4	35# XL gel	25.7%	4.1	58,872
6	B-3C	8.7	Y (12m)	41.4	5	44.7	5	35# XL gel	23.8%	4.3	54,046
7	B-2	19.3	N	25.4	6	26.5	4	35# XL gel	27.2%	4.0	74,149
8	B-1A	14.3	N	34.5	6	36	4.1	35# XL gel	27.3%	3.2	77,729

Figure 3—Modular Tank and Frac tree utilized for improving stage efficiency

As this was the first well in the trial, a DataFRAC was performed prior to every stage. Due to concerns of overflushing, the ball was dropped after flush, averaging 3.3 hrs per stage, with an additional 3.7 hours per stage spent chasing the ball. The pressure during the chasing could not exceed the closure pressure in order not to open the fracture, which added additional time. Five contingency operations from unplanned events represented 75% of the total well time. The utilization of the frac tree improved the pumping cycle efficiency by 36% - by eliminating the need to rig up/rig down the wellhead isolation tool. Overall, the frac stage cycle was 2.05 days, slightly higher than the average frac cycle on the PnP operation (2.0 days). The milling of all the sleeves post job took 14 days. The milling assembly for the plug milling operations was found to be insufficient for the milling the seats and updated assembly and milling procedure was developed.

In order to minimize the risk of gel loading and drastic changes in the additive concentrations, the gel loading optimization was suspended for the following MSF wells. The rate in the M formation was to be maximized, to ensure sufficient frac width was created. The ball drop procedure and contingency plans to for breakdown were updated based on the learnings from this well

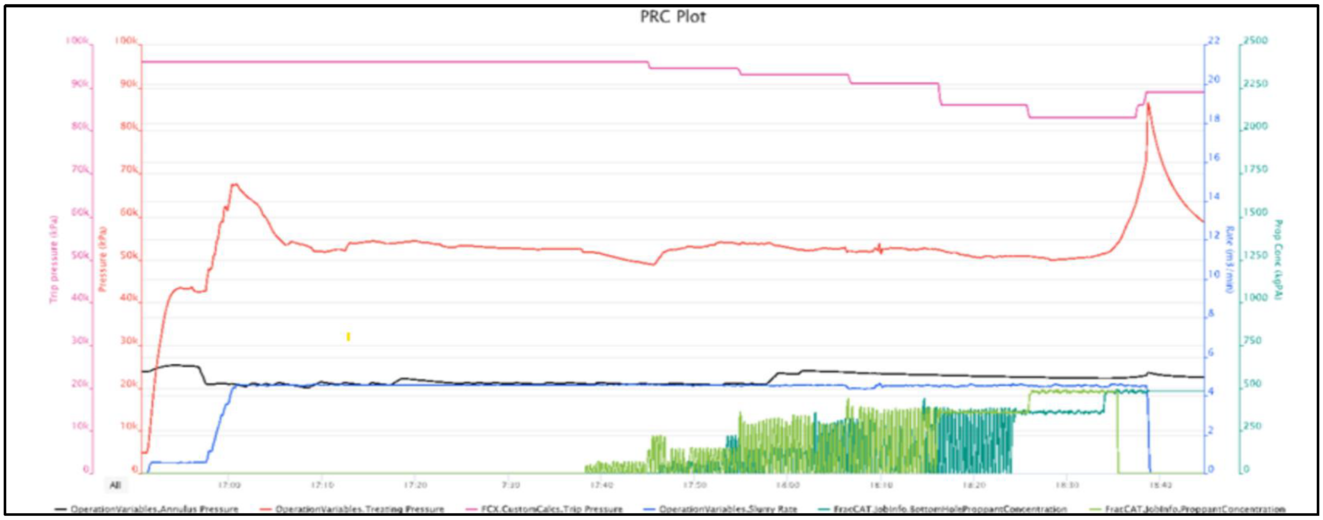

Figure 4—Near wellbore screenout that occurred during stage 4 in Well A, when 4 PPA reached the formation

Well B

Well B was located in the same field, approximately 1190m from Well A. This well was planned to be completed with 7 stages with single port sleeves - 3 sleeves in the M Reservoir and 4 in the B reservoir. The integrity test and opening of the TIV were performed 53 days prior to the start of the frac phase, and the frac tree was installed shortly after. Well B was expected to have similar in-situ stresses as Well A, with the exception of M-3&4. Learning from the previous well, the ball was planned to be dropped during flush to improve the ball seating time and overall efficiency. The stimulation summary is provided in Table 2. Stage 1 saw high NWB friction of 5170 psi. This increased the time to target rate as crosslinked gel had to be gradually introduced as slugs until the pressure allowed for the XL Pad. A sharp increase in pressure was observed when proppant reached the formation, causing a screenout after only 2.5 tons pumped. Due to the stage seat diameter restriction, CT was only able to cleanout until stage 2 seat and the well was kicked off with N2, and flowback was performed until injectivity was recovered. The second attempt introducing 100 mesh sand slugs was unsuccessful too, with the well pressuring out after the third slug (0.75 PPA). The stage was perforated and the target placement of 55t was achieved on the third attempt. Stages 2, 3 and 5 faced issues with breakdown, with wireline perforations being deployed again in Stage 3. The extended duration of the multiple CT cleanout and wireline runs required multiple balls to be dropped during stages 2, 3 & 5 to ensure that the seal was maintained. Due to the low expected potential from B-1 & 2, the sleeves of stages 6 & 7 were opened without fracking.

The improvements in the milling assembly based on learnings from Well A allowed for the milling to be completed in a single run, averaging 2.8 hours per seat – a seven-fold improvement in milling time compared to Well A. The utilization of the frac tree also improved the pump cycle efficiency by 47% compared to PnP operations. While dropping the ball during flush was effective, issues with breakdown and establishing injectivity reduced the efficiency of the operations. Contingency operations were performed in 4 stages – representing 90% of the total time. This increased the stage cycle to 4.84 days, a significant deterioration compared to Well A.

Table 2—Well B Stimulation Summary

Stage	Zone	Distance	Contingency Perf?	Frac Plan (MT)	Max PPA plan	Actual placed (MT)	Max PPA actual	Fluids	Actual PAD%	Avg Rate (m3/min)	Avg Frac Press (kPa)
1	M-5 (TIV)		Y (6m)	55	4.5	60	4.5	40# XL gel	26.2%	4.3	62,499
2	M-3&4	20.5	N	123	6	135	6	40# XL gel	25%	4.4	63,558
3	M-1&2	28.5	Y (6m)	68.4	5	93	5	40# XL gel	24.5%	4.6	62,761
4	B-4	389	N	41	4	52.6	4	35# XL gel	35%	3.9	52,790
5	B-3C	12	N	47	4	52	4	35# XL gel	51%	4.3	62,480
6	B-2	20	N	Sleeve opened only							
7	B-1	15.5	N	Sleeve opened only							

It is likely the remaining cutting and debris from the installation phase affected the injectivity in the TIV. One of the learnings for the upcoming wells was to use it below the pay zone for establishing injectivity to drop the ball for the first stage. Another key learning was the loss of injectivity after wireline perforations. 1-11/16" Strip guns were used for perforation in Stages 1 & 3 due to sleeve seat size restrictions, which generated significant debris. Due to the short rathole volume between the sleeve and seat, the debris (Figure 5) plugged the sleeve ports causing a total loss of injectivity. The sleeve design was revised to eliminate the need for strip guns.

Figure 5—Debris recovered from Coiled Tubing cleanout after perforation run, causing a loss of injectivity in the single-entry sleeves

Well C

Well C was a candidate in a different field, still targeting the B and M formations. The well was designed with 7 single port sleeves, 2 in the M formation and 5 in the B formation. There was an additional PnP stage added to B1 zone. Similar to well B, the balls were dropped during flush after 10-15 bbls, with a target to minimize the ball pumping time to less than 15 seconds. The stimulation summary is provided in Table 3. Overall breakdown was achieved in all stages in the first attempt, except stage 3 & 4 – likely due to the sleeve placement. Treatment in these stages was aborted. A clear opening signature was observed in all stages except stage 4. Injectivity was lost after breakdown in stage 2 and 6, after pumping was stopped due to surface equipment maintenance. As a result, CT cleanout was required for these stages. This further reinforced the need for continuous pumping in single port sleeve systems. High pressure was observed in stage 7, causing the job to be cut short. Overall, 94% placement was achieved with 376 t pumped in 6 stages.

Table 3—Well C Stimulation Summary

Stage	Zone	Distance (m)	Contingency Perf?	Frac Plan (MT)	Max PPA plan	Actual placed (MT)	Max PPA actual	Fluids	Actual PAD%	Avg Rate (m3/min)	Avg Frac Press (kPa)
1	M-4 (TIV)		N	92.6	5	106.5	4.5	40# XL gel	29.9%	4.87	58,862
2	M-3	15	N	66.5	5	76.5	6	40# XL gel	30.1%	3.07	54,058
3	B-9	355	N	21.3	5	Stage skipped due to failed attempts for breakdown					
4	B-8	19	N	30.9	4	Stage skipped due to failed attempts for breakdown					
5	B-7	13	N	66.4	5	76.4	5	35# XL gel	30.0%	2.24	45,577
6	B-6	13	Y	44.1	5	44.3	4	35# XL gel	30.9%	2.77	75,863
7	B-3&4	19	N	25	5	5.3	1.5	35# XL gel	57.0%	2.7	91,971
8	B-1	Perf		55	5	67	5	35# XL gel	30.6%	3.58	67,000

The implementation of the learnings from past wells help bring the stage cycle below PnP operations, to 1.37 days. The milling was performed in 2 runs over 3 days. The first run to mill out the plug for Stage 8, and the second to mill out the seats. The was an increase in milling time over the seats due to restrictions, and accumulation of debris from the previous seats.

The challenges with breakdown and the tracer results were the key drivers in the shifting to multi-entry port systems. Stages 6&7 had decreasing net pay thickness, consequently showing higher pressures. Stage 7 (BK-3/4) had a net pay of 3 m bound by shales and it without another injection point, the fracture would remain localized to the port.

Well D

Well D was completed with a multi-entry sleeves, 2 stages in the M formation and 4 stage in B formation. Stage 2,4, and 6 were completed with 2 ports, Stage 5 with 3 ports. Stage 1 was across the TIV and Stage 3 across B-7 was completed with a single-entry port due to short net pay. Learning from past wells, the aim was to pump continuously, utilizing channel fracturing at the beginning of the treatment to help with placement, and applying a limit to the proppant concentration of 4.5 PPA. Reducing the time between stages, implement a max proppant concentration limit of 4.5 PPA, six stages were completed in 16.5 hours- achieving 108% average proppant placement for the well. The well stimulation summary is provided in Table 4. The frac stage cycle was reduced to 0.1 days.

Table 4—Well D Stimulation Summary

Stage	Zone	Distance (m)	Contingency Perf?	Frac Plan (MT)	Max PPA plan	Actual placed (MT)	Max PPA actual	Fluids	Actual PAD%	Avg Rate (m3/min)	Avg Frac Press (kPa)
1	M-3&4		N	90.3	5	94.8	4	40# XL gel	23.5%	4.0	43,701
2	M-1&2	37	N	90.7	5	95.2	4	40# XL gel	25.4 %	4.4	51,6234
3	B-7	413	N	29.7	5	29.7	4	35# XL gel	22.5 %	3.4	58,835
4	B-5	56	N	75.9	5	91.1	5	35# XL gel	23 %	4.4	42,276
5	B-3&4	50	N	82	5	91	5	35# XL gel	22 %	4.0	48,696
6	B-2&1	26	N	83.2	5	87.3	5	35# XL gel	23 %	4.4	65,657

Deployment summary

The cumulative learnings from each MSF wells showed the improvement in the efficiency over the course of the trials (Figure 6). The stage cycle was reduced by 95% and time between stages was reduced by 99%. Milling was eliminated from the sequence, reducing the risks in the depleted formation, providing a significant advantage over PnP operations.

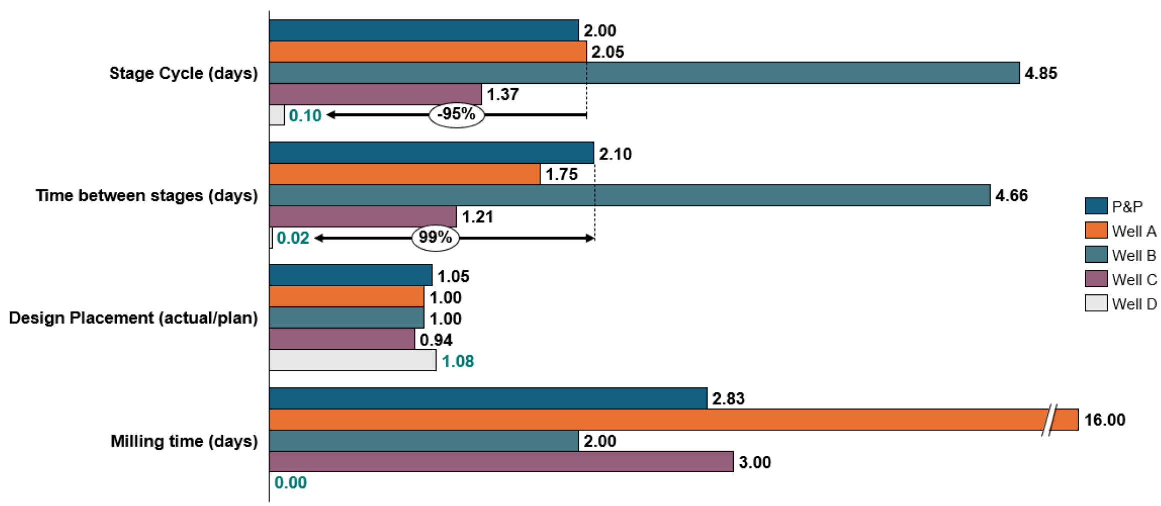

Figure 6—MSF Deployment Summary

Zone coverage/surveillance result

Tracer analysis helped to answer some of the questions related to screenouts observed in the MSF wells. A key observation with the single-entry sleeves was that the frac height coverage was limited when compared against offset wells completed with PnP. The limited height coverage could also explain the screenouts that occurred in Well A, for instance. The differential depletion between zones can further exacerbate this issue. Figure 7 shows the tracer results for Stg 3 in Well A, targeting the M1 and M2 formations. A near wellbore screenout occurred, placing only 41.7 tons in the formation. Tracer analysis showed a height coverage of 6m only, translating to ~7t/m placement across the zone. When comparing the tracer result against the petrophysical data, it become apparent that a small shale streak in middle of M2 is sufficient to arrest the fracture height growth. By comparison a PnP completion typically covers ~18m with 4.5 t/m. Similarly,

Figure 8 show the tracer results for Stage 4 of Well A, covering the B5 formation. A near wellbore screenout occurred after 43.3t were placed, with tracers showing coverage across 8m only, again alluding to a proppant acceptance limit of 7 t/m.

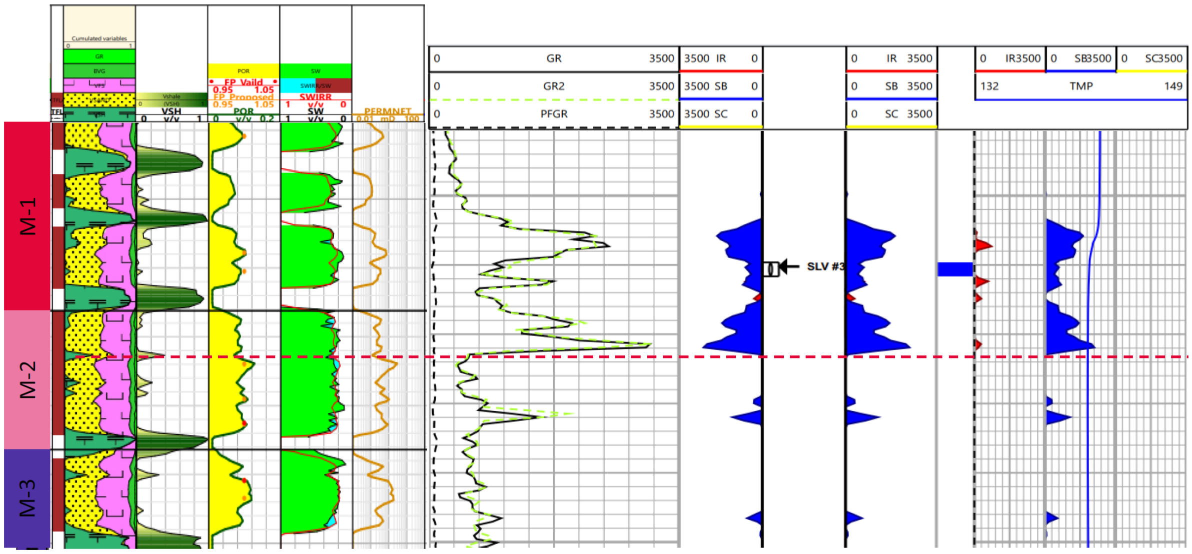

Figure 7—Tracer analysis for Stage 3 in Well A in M-1&2 formation

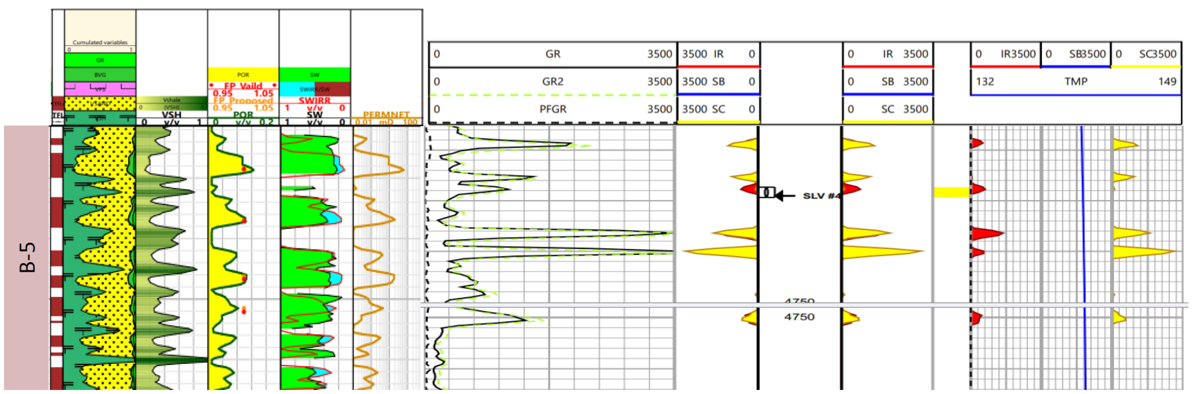

Figure 8—Tracer analysis for Stage 4 in Well A in B-5 formation

In Well B, tracer wash from later stages affected the interpretation of the results. Stage 1 targeting M5 showed adequate coverage across all M5 and some growth into M6. This is likely due to the perforations performed on this stage. Stage 2, which was targeting M3 & M4 showed coverage only across M4, confirming that a single-entry port system would not ensure sufficient coverage across laminated zones. Similarly, in Well C tracer analysis showed inconsistent distribution across the zone, even though there was good cement quality.

As of the time of the writing of this paper, the tracer log for Well D was not yet run, and the surveillance result for this well is pending. It would be crucial to confirm the frac height coverage achieve with the multi-entry ports and compare against results with equivalent PnP completions.

Lessons Learned

Running the completion in vertical wellbores – traditionally this is operationally of less concern vs horizontal wells in terms of drag and stuck pipe risk. But more challenging to achieve the necessary depth accuracy. This is especially true when matching of sleeve port depth to thin layered reservoirs. To verify sleeve's actual depth aligns with proposed depth per open hole log, a CCL ‘tie in’ log is run through the completion once near bottom to confirm.

Operational Efficiency – since the objective of installing an MSF completion is to achieve substantially higher efficiencies; the measure of success becomes a reduction of the total time between frac stages. To this end, full success was achieved after 4 wells. Several factors contributed to this success including.

Opening the sleeve – proven most successful when pumping never stopped between frac stages and opening was confirmed by acoustic monitoring at surface

Modular tank- to store at the wellsite enough water to be immediately available and allow continuous pumping for all stages

High formation breakdown pressures – several zones showed breakdown pressures that exceeded the safe tubing pressure limit. This meant perforations were required. To compensate for this possible outcome the following measures were taken:

Selection of MSF port depth -was picked to be nearer the bottom of the zone to allow space for potential perfs above and still maintain isolation from below during frac

Perforations – since guns needed to pass through all upper stage ball seats the gun size was limited in diameter, which at times meant exposed guns which left debris on top of isolation ball but no detrimental effects were encountered

Limited Entry Sleeves – trialed with mixed results. While it does allow 2+ separate zones to be stimulated simultaneously the maximum allowable proppant concentration through this smaller nozzle/port is less than standard design in conventional wells

Screenout avoidance – To achieve the full operational efficiency in the MSF completion, its paramount to avoid screenouts. Compared to a PnP completion, the sleeve seat restrictions pose access issues and multiple CT cleanout runs could also affect the integrity of the seal, requiring multiple balls to be deployed and placement uncertainty. A more conservative design approach should be applied for MSF completions

Isolation and surveillance – The gravity drop method was inconclusive and inefficient, and dropping the ball during flush ensured a clear confirmation of the ball seating and sleeve opening. It was also found that pumping multiple balls to isolate the same stage due to degradation concerns did not ensure sufficient isolation, as seen by the tracer wash. While tracers from the subsequent stages interfered with the analysis, a reasonable estimation of frac height was still obtained – providing valuable insight into the frac height design

Conclusion

In this paper we share the journey and lessons learned of the application of multistage frac completions in depleted vertical wells, and their impact on the frac placement. We highlight the incremental design changes in each well, and the learnings that were implemented in the subsequent wells to achieve full success after four wells. The modifications on the frac design also helped to prevent near wellbore screenouts and eliminate the need for CT cleanouts. While production from multi-entry sleeve completions is comparable with PnP wells, tracer analysis needs to be completed to confirm frac coverage across the zones.

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